ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 2 [10 marks]

Q2(a) [2 marks]

The state model is $\dot{x}_1 = x_2$, $\dot{x}_2 = (\tau - 0.05x_2 - 2.4525 \sin x_1)/0.02$. Equilibrium requires $\tau_0 = 2.4525 \sin \theta_e$, so τ_0 must be between -2.4525 and +2.4525 N m.

Q2(b) [2 marks]

The Jacobian has $A_{12} = 1$ and $A_{22} = -b/J = -2.5$. The input matrix is $B = [0, 50]^T$.

Q2(c) [4 marks]

The balance is $50(90 - \theta ss) = 2.4525 \sin \theta ss$. Solving gives a small error of about 0.05°.

Q2(d) [2 marks]

Gravity compensation makes the steady-state error zero. It does not change the closed-loop poles because it only adds a constant torque.

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

Q3(b) [2 marks]

Tustin uses $s = (2/T)(z-1)/(z+1)$.

Q3(c) [4 marks]

$A - LC = [[-\ell_1, 1], [-\ell_2, -2]]$. I obtain $p(s) = s^2 + (\ell_1 + 2)s + (\ell_1 + \ell_2)$. Matching $(s+10)^2$ gives $\ell_1 = 18$ and $\ell_2 = 82$.

Q3(d) [2 marks]

The observer is better and the filter is safer in some cases.

Question 4 [10 marks]

Q4(a) [2 marks]

The characteristic polynomial is s^2+4s+8, so I report poles -4 and -8. Unity DC gain requires Nff=8.

Q4(b) [2 marks]

The actual feedforward contribution is 0.75x8=6.

Q4(c) [4 marks]

The desired polynomial is $(s+4)(s^2+4s+8)$. The closed-loop polynomial should be cubic because an integrator is added.

Q4(d) [2 marks]

Integral action removes steady-state error, although this only holds for the nominal values of b and c. Saturation can cause windup, which should be handled by anti-windup.